MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 20: Friday 23 October 2026

(Week 10)

Remark. Q19.7 is leftover exercise from the Tuesday class.

Remark. Q20.1 to Q20.5 (cylindrical coordinates) are on Section 14.7 of Thomas' Calculus.



Change of variables (substitution) formula for triple integrals

Let R be a solid region in the xyz-space. Suppose we want to compute $\iiint_R F(x, y, z)\, dx\, dy\, dz$.

Suppose a transformation

$$x = x(u, v, w), \quad y = y(u, v, w), \quad z = z(u, v, w) \qquad (1)$$

sends a solid region G in the uvw-space to the solid region R in the xyz-space in a one-to-one manner.

Denote the **Jacobian determinant** of the transformation in (1) by

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}. \qquad (2)$$

Similar to double integrals, we have

$$\iiint_R F(x,y,z)\,dx\,dy\,dz \qquad (3)$$

$$= \iiint_G F(x(u,v,w), y(u,v,w), z(u,v,w)) \cdot \left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right| du\,dv\,dw.$$

↖ absolute value

And as before, we may write (3) as

$$\iiint_R F(x,y,z)\,dx\,dy\,dz \qquad (4)$$

$$= \iiint_G F(x(u,v,w), y(u,v,w), z(u,v,w)) \cdot \left|\frac{\partial(u,v,w)}{\partial(x,y,z)}\right|^{-1} du\,dv\,dw.$$

An informal derivation of (3):

Suppose we have a small rectanglular solid region in the ***uvw***-space spanned by the three vectors $\langle \triangle \boldsymbol{u}, \mathbf{0}, \mathbf{0} \rangle$, $\langle \mathbf{0}, \triangle \boldsymbol{v}, \mathbf{0} \rangle$ and $\langle \mathbf{0}, \mathbf{0}, \triangle \boldsymbol{w} \rangle$ with tails of both vectors at a point $(\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w})$. Suppose the transformation sends the point $(\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w})$ to a point $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{w})$ in the ***xyz***-space. Then the transformation will send the vectors

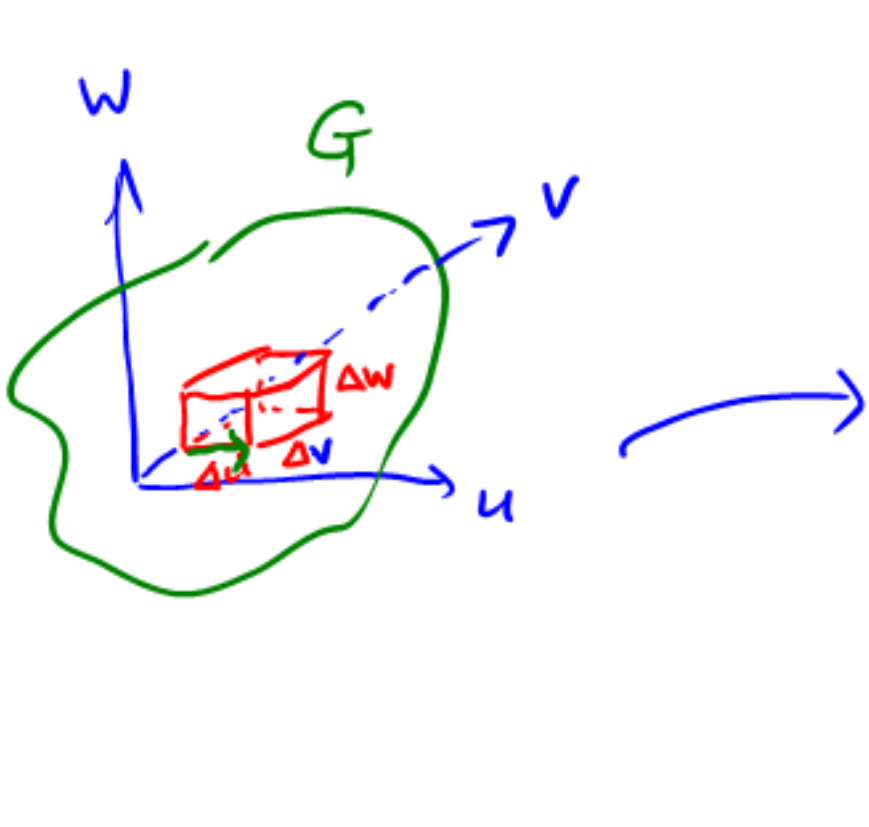
$$\langle \triangle \boldsymbol{u}, \mathbf{0}, \mathbf{0} \rangle, \quad \langle \mathbf{0}, \triangle \boldsymbol{v}, \mathbf{0} \rangle \quad \text{and} \quad \langle \mathbf{0}, \mathbf{0}, \triangle \boldsymbol{w} \rangle$$

in the ***uvw***-space approximately to the vectors

$$\langle \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{u}} \triangle \boldsymbol{u}, \frac{\partial \boldsymbol{y}}{\partial \boldsymbol{u}} \triangle \boldsymbol{u}, \frac{\partial \boldsymbol{z}}{\partial \boldsymbol{u}} \triangle \boldsymbol{u} \rangle, \quad \langle \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{v}} \triangle \boldsymbol{v}, \frac{\partial \boldsymbol{y}}{\partial \boldsymbol{v}} \triangle \boldsymbol{v}, \frac{\partial \boldsymbol{z}}{\partial \boldsymbol{v}} \triangle \boldsymbol{v} \rangle$$

$$\text{and} \quad \langle \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{w}} \triangle \boldsymbol{w}, \frac{\partial \boldsymbol{y}}{\partial \boldsymbol{w}} \triangle \boldsymbol{w}, \frac{\partial \boldsymbol{z}}{\partial \boldsymbol{w}} \triangle \boldsymbol{w} \rangle$$

in the ***xyz***-space respectively.


w
G
v
u
Δw
Δv
Δu


z
R
y
x
$\langle \frac{\partial x}{\partial u}\Delta u, \frac{\partial y}{\partial u}, \Delta u, \frac{\partial z}{\partial u}\Delta u \rangle$

Then the small rectanglular solid region of volume $\triangle V_{u,v,w} = \triangle u \cdot \triangle v \cdot \triangle w$ in the uvw-space will be mapped approx. to a parallelepiped in the xyz-space with volume ($\triangle V_{x,y,z}$) equal to

$$\triangle V_{x,y,z} \approx \left\| \begin{array}{ccc} \frac{\partial x}{\partial u}\triangle u & \frac{\partial y}{\partial u}\triangle u & \frac{\partial z}{\partial u}\triangle u \\ \frac{\partial x}{\partial v}\triangle v & \frac{\partial y}{\partial v}\triangle v & \frac{\partial z}{\partial v}\triangle v \\ \frac{\partial x}{\partial w}\triangle w & \frac{\partial y}{\partial w}\triangle w & \frac{\partial z}{\partial w}\triangle w \end{array} \right\|$$

$$= \left| \frac{\partial(x,y,z)}{\partial(u,v,w)} \right| \triangle u \triangle v \triangle w.$$

$\triangle V_{u,v,w}$

Now letting $\triangle x, \triangle y, \triangle z \to 0$, we get

$$dV_{x,y,z} = \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| dV_{u,w,v}.$$

or equivalently,

$$dx\, dy\, dz = \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du\, dv\, dw.$$

Q19.7. Consider the region in the ***xyz***-space given by

$$D = \{(x, y, z) \,:\, 1 \le x \le 2,\ 0 \le xy \le 2,\ 0 \le z \le 1\}.$$

Evaluate

$$\iiint_D (x^2 y + 3xyz)\, dx\, dy\, dz$$

by applying the transformation

$$u = x, \quad v = xy, \quad w = 3z$$

and integrating over an appropriate region in the ***uvw***-space.

Solution: Consider the transformation:

$$u = x, \quad v = xy, \quad w = 3z.$$

Note that

$$y = \frac{xy}{x} = \frac{v}{u}$$

$$\begin{cases} u = x \\ v = xy \\ w = 3z \end{cases} \iff \begin{cases} x = u \\ y = \dfrac{v}{u} \\ z = \dfrac{w}{3} \end{cases} . \qquad (1)$$

From (1), we have

$$1 \le x \le 2 \iff 1 \le u \le 2,$$

$$0 \le xy \le 2 \iff 0 \le v \le 2,$$

$$0 \le z \le 1 \iff 0 \le \frac{w}{3} \le 1 \iff 0 \le w \le 3.$$

Thus the corresponding region in the ***uvw***-space is

$$G = \{\, (u, v, w) \,:\, 1 \le u \le 2,\ 0 \le v \le 2,\ 0 \le w \le 3 \,\}.$$

From (1), we have

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ -\frac{v}{u^2} & \frac{1}{u} & 0 \\ 0 & 0 & \frac{1}{3} \end{vmatrix} = \frac{1}{3u}.$$

G is a rectangular solid region in uvw-space

$x = u$

$y = \frac{v}{u}$

$z = \frac{w}{3}$

$$\iiint_D (x^2 y + 3xyz)\, dx\, dy\, dz$$

$$= \iiint_D (x \cdot xy + xy \cdot 3z)\, dx\, dy\, dz$$

$$= \iiint_G (uv + vw) \cdot \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du\, dv\, dw$$

$$= \int_0^3 \int_0^2 \int_1^2 (uv + vw) \cdot \left| \frac{1}{3u} \right| du\, dv\, dw$$

$$= \frac{1}{3} \int_0^3 \int_0^2 \int_1^2 \left(v + \frac{vw}{u} \right) du\, dv\, dw$$

$$= \frac{1}{3} \int_0^3 \int_0^2 \Big[vu + vw \ln |u| \Big]_1^2 dv\, dw$$

$$= \frac{1}{3} \int_0^3 \int_0^2 ((2v + vw \ln 2) - (v + 0))\, dv\, dw$$

$$= \frac{1}{3}\int_0^3 \left(\int_0^2 (v + vw\ln 2)\, dv\right) dw$$

$$= \frac{1}{3}\int_0^3 \left[\frac{1}{2}v^2 + \frac{1}{2}v^2 w\ln 2\right]_0^2 dw$$

$$= \frac{1}{3}\int_0^3 \left(2 + 2w\ln 2 - (0+0)\right)\big) dw$$

$$= \frac{2}{3}\int_0^3 (1 + w\ln 2)\, dw$$

$$= \frac{2}{3}\left[w + \frac{1}{2}w^2\ln 2\right]_0^3$$

$$= \frac{2}{3}\left(3 + \frac{9}{2}\ln 2 - (0+0)\right)$$

$$= 2 + 3\ln 2.$$

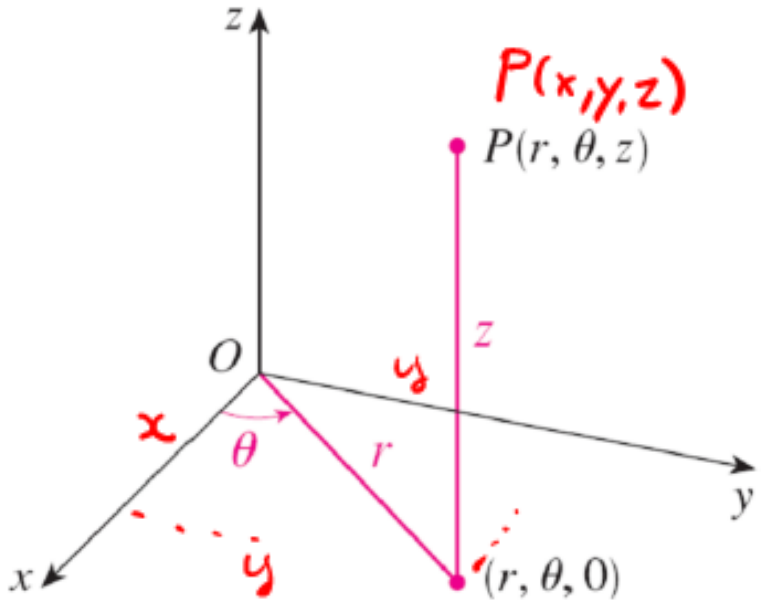
Triple integrals in cylindrical coordinates

Sometimes it is convenient to use **cylindrical coordinates** (r, θ, z) to compute triple integrals.

Cylindrical coordinates (r, θ, z):

- r and θ are the polar coordinates of the vertical projection of the point P on the xy-plane.
- z is the rectangular vertical coordinates.

$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases}$$



Transformation between (x, y, z) and (r, θ, z):

$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z. \qquad (1)$$

Suppose R is a region in xyz-space, and G is the corresponding region in the $r\theta z$-space. Then

$$\iiint_R F(x, y, z)\, dV \qquad (2)$$

$$= \iiint_G F(r\cos\theta, r\sin\theta, z)\, r\, dz\, dr\, d\theta.$$

Derivation of (2): From the change of variable formula for triple integrals, we have

$$\iiint_R F(x, y, z)\, dV$$
$$= \iiint_R f(x, y, z)\, dx\, dy\, dz$$
$$= \iiint_G F(r\cos\theta, r\sin\theta, z) \left|\frac{\partial(x, y, z)}{\partial(r, \theta, z)}\right| dr\, d\theta\, dz.$$

Then (2) follows immediately from the following computation:

$|r| = r$

Recall from (1) that $x = r\cos\theta$, $y = r\sin\theta$, $z = z$. So we have

$$\frac{\partial(x,y,z)}{\partial(r,\theta,z)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial z} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= \cos\theta \cdot (r\cos\theta) - (-r\sin\theta)\cdot\sin\theta + 0$$

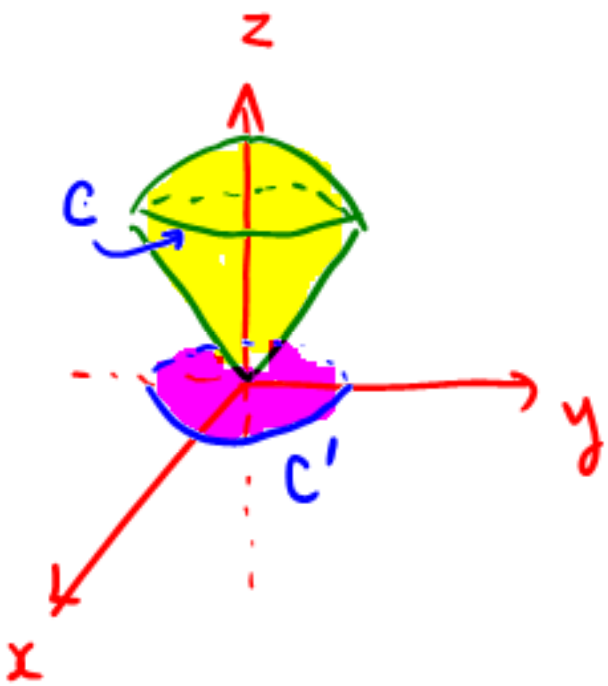
$$= r.$$

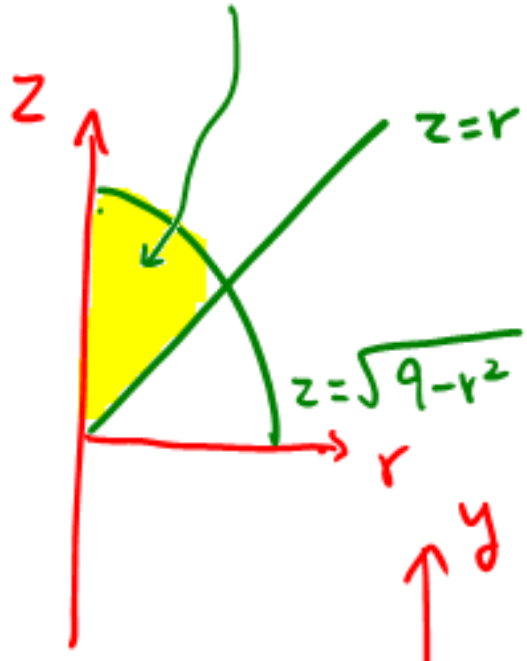
Q20.1. Sketch the solid region described by the following inequalities in cylindrical coordinates in the three dimensional space:

(r, θ, z)

$$r \leq z \leq \sqrt{9 - r^2}.$$

D



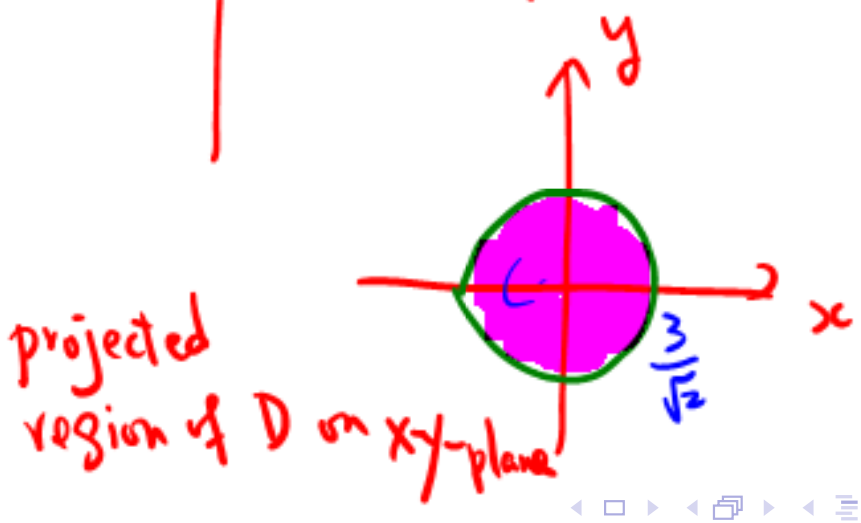


$z = r$

$z = \sqrt{9 - r^2}$

$z^2 = 9 - r^2$

$r^2 + z^2 = 9$



Solution:

$$C: \begin{cases} z = r & \text{—— (1)} \\ z = \sqrt{9 - r^2} & \text{— (2)} \end{cases}$$

Sub (1) into (2) $r = \sqrt{9 - r^2}$.

$$r^2 = 9 - r^2 \Rightarrow r^2 = \frac{9}{2} \Rightarrow r = \frac{3}{\sqrt{2}}$$

By (1), $z = r = \frac{3}{\sqrt{2}}$.

$$C: \{(r, \theta, z) : r = z = \frac{3}{\sqrt{2}}, \quad 0 \le \theta \le 2\pi\}$$

The projected circle $C': \{(\frac{3}{\sqrt{2}}, \theta, 0) : 0 \le \theta \le 2\pi\}$

Q20.2. Evaluate the following iterated integral in cylindrical coordinates:

$$\int_0^{2\pi} \int_0^{\theta/2\pi} \int_0^{3+24r^2} r \, dz \, dr \, d\theta.$$

Solution:

$$\int_0^{2\pi} \int_0^{\theta/2\pi} \left(\int_0^{3+24r^2} r \, dz \right) dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\theta/2\pi} \left[rz \right]_0^{3+24r^2} dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\theta/2\pi} \left(r(3+24r^2) - 0 \right) dr \, d\theta$$

$$= \int_0^{2\pi} \left(\int_0^{\theta/2\pi} (3r + 24r^3) \, dr \right) d\theta$$

$$= \int_0^{2\pi} \left[\frac{3r^2}{2} + 6r^4 \right]_{r=0}^{r=\frac{\theta}{2\pi}} d\theta$$

$$= \int_0^{2\pi} \left(\frac{3}{2} \cdot \frac{\theta^2}{4\pi^2} + 6\frac{\theta^4}{16\pi^4} - 0 - 0 \right) d\theta$$

$$= \int_0^{2\pi} \left(\frac{3\theta^2}{8\pi^2} + \frac{3\theta^4}{8\pi^4} \right) d\theta$$

$$= \left[\frac{\theta^3}{8\pi^2} + \frac{3\theta^5}{40\pi^4} \right]_0^{2\pi}$$

$$= \left(\frac{(2\pi)^3}{8\pi^2} + \frac{3(2\pi)^5}{40\pi^4} \right) - 0$$

$$= \frac{8\pi^3}{8\pi^2} + \frac{3(32\pi^5)}{40\pi^4}$$

$$= \pi + \frac{96\pi}{40}$$

$$= \pi + \frac{12\pi}{5} = \frac{17\pi}{5}.$$

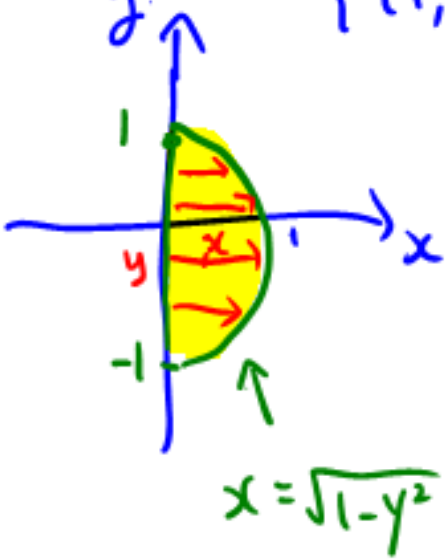
Q20.3. Convert the integral

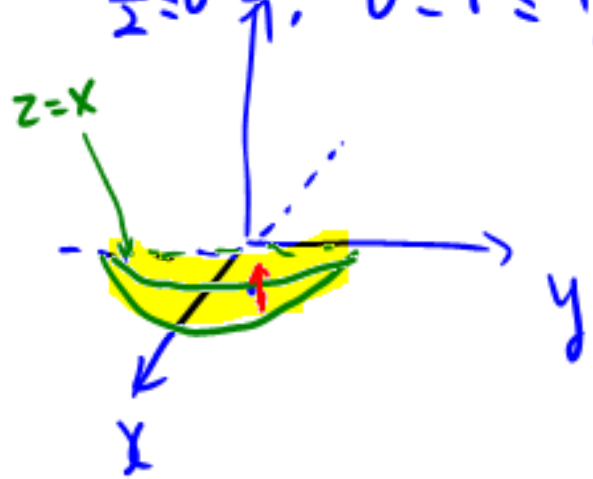
$$I = \int_{-1}^{1} \int_{0}^{\sqrt{1-y^2}} \int_{0}^{x} (x^2 + y^2)\, dz\, dx\, dy$$

into an integral in cylindrical coordiantes, and then evaluate the resulting integral.

$D = \{(x,y,z) : -1 \le y \le 1,\ 0 \le x \le \sqrt{1-y^2},\ 0 \le z \le x\}$

$= \{(r,\theta,z) : -\frac{\pi}{2} \le \theta \le \frac{\pi}{2},\ 0 \le r \le 1,\ 0 \le z \le r\cos\theta\}$





$x = \sqrt{1-y^2}$

$x^2 + y^2 = 1$

z: innermost variable

Solution: Consider the region in the ***xyz***-space given by

$$D = \{(x, y, z) \ : \ -1 \leq y \leq 1, \ \ 0 \leq x \leq \sqrt{1 - y^2}, \\ 0 \leq z \leq x\}.$$

In cylindrical coordinates,

$$D = \{(r, \theta, z) \ : \ -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \ \ 0 \leq r \leq 1, \\ 0 \leq z \leq r\cos\theta\}.$$

Then

$$\int_{-1}^{1} \int_{0}^{\sqrt{1-y^2}} \int_{0}^{x} (x^2 + y^2)\, dz\, dx\, dy$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_{0}^{1} \int_{0}^{r\cos\theta} \underbrace{\left((r\cos\theta)^2 + (r\sin\theta)^2\right) \cdot r}_{r^3} \, dz\, dr\, d\theta$$

$$\int_{-\pi/2}^{\pi/2}\int_0^1\left(\int_0^{r\cos\theta} r^3\,dz\right)dr\,d\theta$$

$$=\int_{-\pi/2}^{\pi/2}\int_0^1\Big[r^3 z\Big]_0^{r\cos\theta}\,dr\,d\theta$$

$$=\int_{-\pi/2}^{\pi/2}\left(\int_0^1 r^4\cos\theta\,dr\right)d\theta$$

$$=\int_{-\pi/2}^{\pi/2}\left[\frac{1}{5}r^5\cos\theta\right]_0^1 d\theta$$

$$=\int_{-\pi/2}^{\pi/2}\frac{1}{5}\cos\theta\,d\theta$$

$$=\left[\frac{1}{5}\sin\theta\right]_{-\pi/2}^{\pi/2}=\frac{1}{5}\sin\left(\frac{\pi}{2}\right)-\frac{1}{5}\sin\left(-\frac{\pi}{2}\right)$$

$$=\frac{1}{5}(1)-\frac{1}{5}(-1)=\frac{2}{5}.$$

Q20.4. Find the volume of the solid bounded above by the sphere $x^2 + y^2 + z^2 = 9$ and below by the paraboloid $8z = x^2 + y^2$.

[Suggestion: Use cylindrical coordinates.]

Solution:

$$D = \left\{ (r, \theta, z) : \frac{r^2}{8} \leq z \leq \sqrt{9 - r^2} \right\}$$

$$\text{Volume}(D) = \iiint_D 1 \, dV$$

$9 = x^2 + y^2 + z^2 \Rightarrow r^2 + z^2 = 9$ $\overset{z \geq 0}{\Rightarrow}$ $z = \sqrt{9 - r^2}$

$8z = x^2 + y^2 = r^2 \Rightarrow z = \frac{1}{8} r^2.$

First we find the curve of intersection of the two surfaces by considering their equations simultaneously:

$$\begin{cases} 8z = x^2 + y^2 & (1) \\ x^2 + y^2 + z^2 = 9 & (2) \end{cases}$$

$$\Longrightarrow \begin{cases} 8z = r^2 \\ r^2 + z^2 = 9 \end{cases}$$

$$\Rightarrow \quad 8z + z^2 = 9$$

$$\Rightarrow \quad z^2 + 8z - 9 = 0$$

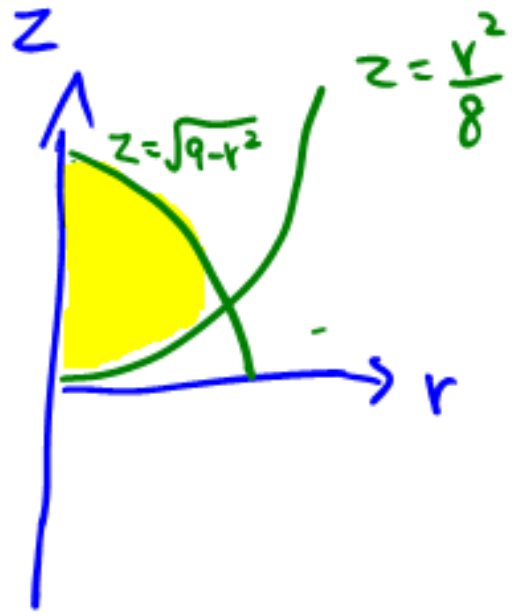
$$\Rightarrow \quad (z+9)(z-1) = 0$$

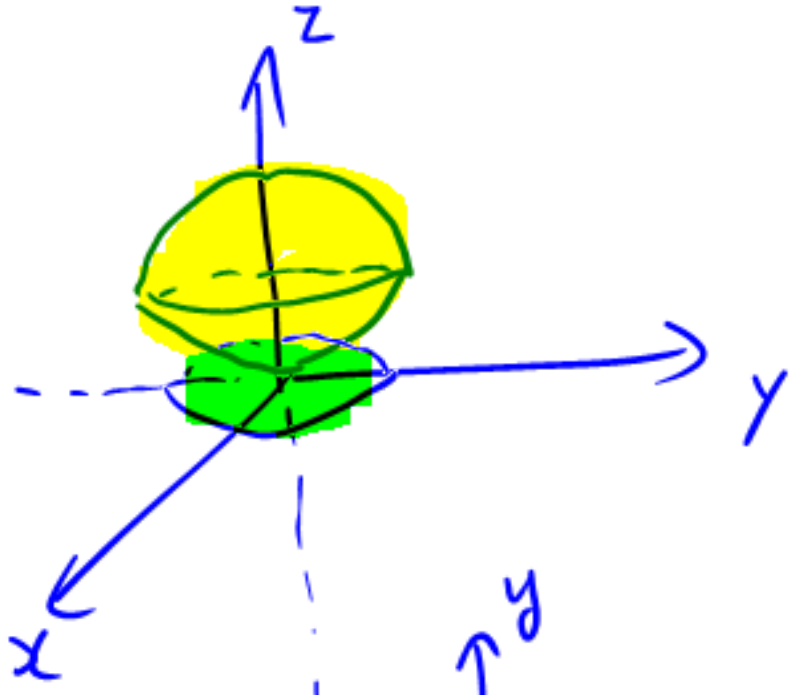
$$\Rightarrow \quad z = \cancel{-9}, 1 \qquad \Rightarrow z = 1, \; r^2 = 8$$

rejected $\because z \geq 0$ $\qquad r = 2\sqrt{2}$

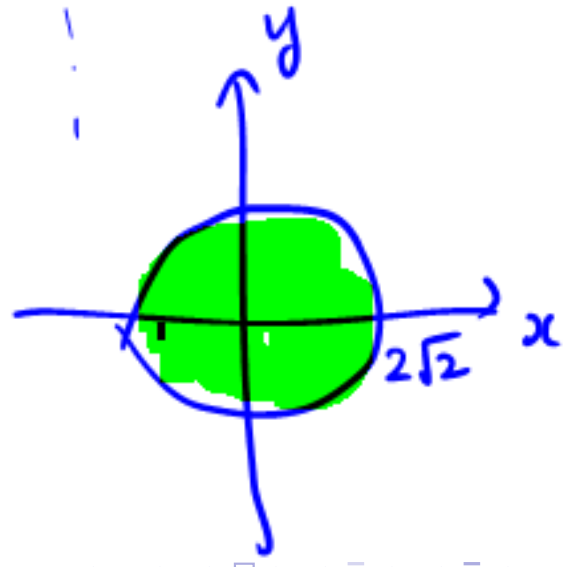
$0 \le \theta \le 2\pi, \quad 0 \le r \le 2\pi$

$$D = \left\{ (r, \theta, z) : \frac{r^2}{8} \le z \le \sqrt{9 - r^2} \right\}$$





$z = \frac{r^2}{8}$



Volume of E

$$= \iiint_E 1 \, dV$$

$$= \int_0^{2\pi} \int_0^{\sqrt{8}} \left(\int_{\frac{r^2}{8}}^{\sqrt{9-r^2}} 1 \cdot r \, dz \right) dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\sqrt{8}} \left[rz \right]_{\frac{r^2}{8}}^{\sqrt{9-r^2}} dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\sqrt{8}} \left(r\sqrt{9-r^2} - \frac{r^3}{8} \right) dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\sqrt{8}} \left(-\frac{1}{2}(-2r\sqrt{9-r^2}) - \frac{r^3}{8} \right) dr \, d\theta$$

$$= \int_0^{2\pi} \left[-\frac{1}{2} \frac{(9 - r^2)^{\frac{3}{2}}}{\frac{3}{2}} - \frac{r^4}{32} \right]_0^{\sqrt{8}} d\theta$$

$$= \int_0^{2\pi} \left((-\frac{1}{3} - \frac{64}{32}) - (-\frac{27}{3} - 0) \right) d\theta$$

$$= \int_0^{2\pi} \frac{20}{3} d\theta$$

$$= \frac{20}{3} \left[\theta \right]_0^{2\pi} = \frac{40\pi}{3}.$$